

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    18:17:19 04/12/2011
6  -- Design Name:
7  -- Module Name:    reg2bit - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity reg2bit is
24     Port ( clk : in  STD_LOGIC;
25           clr : in  STD_LOGIC;
26           inp : in  STD_LOGIC_VECTOR (1 downto 0);
27           load : in  STD_LOGIC;
28           q : inout STD_LOGIC_VECTOR (1 downto 0));
29 end reg2bit;
30
31 architecture Behavioral of reg2bit is
32
33     component reg1bit is
34         Port ( clr : in  STD_LOGIC;
35               clk : in  STD_LOGIC;
36               load : in  STD_LOGIC;
37               inp : in  STD_LOGIC;
38               q : inout STD_LOGIC);
39     end component;
40
41     begin
42
43     U1 : reg1bit port map( clk => clk, clr =>clr, inp => inp(0), load => load, q => q(0)
44         );
45
46     U2 : reg1bit port map( clk => clk, clr =>clr, inp => inp(1), load => load, q => q(1)
47         );
48 end Behavioral;
49
50
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   18:22:27 04/12/2011
6  -- Design Name:
7  -- Module Name:   F:/Senior Project/reg2bit/reg2bit_TB.vhd
8  -- Project Name:  reg2bit
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: reg2bit
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32  ENTITY reg2bit_TB IS
33  END reg2bit_TB;
34
35  ARCHITECTURE behavior OF reg2bit_TB IS
36
37      -- Component Declaration for the Unit Under Test (UUT)
38
39      COMPONENT reg2bit
40      PORT(
41          clk : IN  std_logic;
42          clr : IN  std_logic;
43          inp : IN  std_logic_vector(1 downto 0);
44          load : IN  std_logic;
45          q : INOUT std_logic_vector(1 downto 0)
46      );
47      END COMPONENT;
48
49
50      --Inputs
51      signal clk : std_logic := '0';
52      signal clr : std_logic := '0';
53      signal inp : std_logic_vector(1 downto 0) := (others => '0');
54      signal load : std_logic := '0';
55
56      --BiDirs
```

```
57     signal q : std_logic_vector(1 downto 0);
58
59     -- Clock period definitions
60     constant clk_period : time := 10 ns;
61
62 BEGIN
63
64     -- Instantiate the Unit Under Test (UUT)
65     uut: reg2bit PORT MAP (
66         clk => clk,
67         clr => clr,
68         inp => inp,
69         load => load,
70         q => q
71     );
72
73     -- Clock process definitions
74     clk_process : process
75     begin
76         clk <= '0';
77         wait for clk_period/2;
78         clk <= '1';
79         wait for clk_period/2;
80     end process;
81
82
83     --load process
84     load_process: process
85     begin
86         load <= '0';
87         wait for 40 ns;
88         load <= '1';
89         wait for 20 ns;
90         load <= '0';
91         wait for 40 ns;
92     end process;
93
94     --input process
95     inp_process: process
96     begin
97         inp <= "01";
98         for i in 0 to 2 loop
99             wait for 100 ns;
100             inp <= std_logic_vector(unsigned(inp)+1);
101         end loop;
102     end process;
103
104
105
106     -- Stimulus process
107     stim_proc: process
108     begin
109         clr <= '1' ;
110         wait for 10 ns;
111         clr <= '0';
112         wait for 10 ns;
```

```
113
114     wait;
115     end process;
116
117     END;
118
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    18:34:45 04/12/2011
6  -- Design Name:
7  -- Module Name:    reg2bit_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity reg2bit_top is
24     Port ( mclk : in  STD_LOGIC;
25           sw  : in  STD_LOGIC_VECTOR (1 downto 0);
26           btn : in  STD_LOGIC_VECTOR (3 downto 2);
27           ld  : inout STD_LOGIC_VECTOR (1 downto 0));
28 end reg2bit_top;
29
30 architecture Behavioral of reg2bit_top is
31 component reg2bit is
32     port(  clk : in  STD_LOGIC;
33           clr : in  STD_LOGIC;
34           inp : in  STD_LOGIC_VECTOR (1 downto 0);
35           load : in  STD_LOGIC;
36           q  : inout STD_LOGIC_VECTOR (1 downto 0));
37 end component;
38
39 begin
40
41 U1 : reg2bit port map( clk => mclk, clr => btn(3), inp => sw, load => btn(2), q =>
42 ld);
43
44 end Behavioral;
45
```

```
1  NET "sw[0]" LOC = G18;
2  NET "sw[1]" LOC = H18;
3  NET "btn[2]" LOC = E18;
4  NET "btn[3]" LOC = H13;
5  NET "ld[0]" LOC = J14;
6  NET "ld[1]" LOC = J15;
7  NET "mclk" LOC = B8;
8
9
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    20:13:51 04/05/2011
6  -- Design Name:
7  -- Module Name:    reg - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity reg is
24     generic(N: integer := 8);
25     Port ( load : in  STD_LOGIC;
26           clk  : in  STD_LOGIC;
27           clr  : in  STD_LOGIC;
28           d    : in  STD_LOGIC_VECTOR (N-1 downto 0);
29           q    : out STD_LOGIC_VECTOR (N-1 downto 0));
30 end reg;
31
32 architecture Behavioral of reg is
33
34 begin
35     process(clk, clr)
36     begin
37         if clr = '1' then
38             q <= (others => '0');
39         elsif clk'event and clk = '1' then
40             if load = '1' then
41                 q <= D;
42             end if;
43         end if;
44     end process;
45 end Behavioral;
46
47
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    20:17:52 04/05/2011
6  -- Design Name:
7  -- Module Name:    F:/Senior Project/register/reg_TB.vhd
8  -- Project Name:   register
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: reg
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32  ENTITY reg_TB IS
33  END reg_TB;
34
35  ARCHITECTURE behavior OF reg_TB IS
36
37      -- Component Declaration for the Unit Under Test (UUT)
38
39      COMPONENT reg
40      PORT(
41          load : IN  std_logic;
42          clk  : IN  std_logic;
43          clr  : IN  std_logic;
44          d    : IN  std_logic_vector(7 downto 0);
45          q    : OUT std_logic_vector(7 downto 0)
46      );
47      END COMPONENT;
48
49
50      --Inputs
51      signal clk : std_logic := '0';
52      signal clr : std_logic := '0';
53      signal d   : std_logic_vector(7 downto 0) := (others => '0');
54      signal load : std_logic := '0';
55
56      --Outputs
```



```
57     signal q : std_logic_vector(7 downto 0);
58
59     -- Clock period definitions
60     constant clk_period : time := 10 ns;
61
62 BEGIN
63
64     -- Instantiate the Unit Under Test (UUT)
65     uut: reg PORT MAP (
66         load => load,
67         clk => clk,
68         clr => clr,
69         d => d,
70         q => q
71     );
72
73     -- Clock process definitions
74     clk_process :process
75     begin
76         clk <= '0';
77         wait for clk_period/2;
78         clk <= '1';
79         wait for clk_period/2;
80     end process;
81
82
83     --load process
84     load_process: process
85     begin
86         load <= '0';
87         wait for 40 ns;
88         load <= '1';
89         wait for 20 ns;
90         load <= '0';
91         wait for 40 ns;
92     end process;
93
94     --input process
95     d_process: process
96     begin
97         d <= X"3A";
98         wait for 75 ns;
99         d <= X"8F";
100        wait for 100 ns;
101        d <= X"C7";
102        wait for 200 ns;
103    end process;
104
105
106
107     -- Stimulus process
108     stim_proc: process
109     begin
110         clr <= '1' ;
111         wait for 10 ns;
112         clr <= '0';
```

```
113         wait for 10 ns;
114
115         wait;
116     end process;
117
118 END;
119
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    16:11:12 04/08/2011
6  -- Design Name:
7  -- Module Name:    register_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity register_top is
24     Port ( mclk : in  STD_LOGIC;
25           sw : in  STD_LOGIC_VECTOR (7 downto 0);
26           btn : in  STD_LOGIC_VECTOR (3 downto 2);
27           ld : out STD_LOGIC_VECTOR (7 downto 0);
28           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
29           an : out STD_LOGIC_VECTOR (3 downto 0);
30           dp : out STD_LOGIC);
31 end register_top;
32
33 architecture Behavioral of register_top is
34 component clkdiv is
35     Port ( mclk : in  STD_LOGIC;
36           clr : in  STD_LOGIC;
37           clk190 : out STD_LOGIC
38           );
39 end component;
40
41 component reg is
42     generic(N: integer := 8);
43     Port ( load : in  STD_LOGIC;
44           clk : in  STD_LOGIC;
45           clr : in  STD_LOGIC;
46           d : in  STD_LOGIC_VECTOR (N-1 downto 0);
47           q : out STD_LOGIC_VECTOR (N-1 downto 0)
48           );
49 end component;
50
51 component x7segb is
52     Port ( x : in  STD_LOGIC_VECTOR (15 downto 0);
53           clk : in  STD_LOGIC;
54           clr : in  STD_LOGIC;
55           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
56           an : out  STD_LOGIC_VECTOR (3 downto 0);
```

```
57         dp : out   STD_LOGIC
58     );
59 end component;
60
61 signal clk190 : STD_LOGIC;
62 signal g, q : STD_LOGIC_VECTOR(7 downto 0);
63 signal x : STD_LOGIC_VECTOR(15 downto 0);
64
65 begin
66
67     g <= "00000000";
68
69     U1 : clkdiv
70         port map( mclk => mclk, clr => btn(3), clk190 => clk190);
71
72     U2 : reg
73         generic map ( N=> 8)
74         port map( clk => mclk, clr => btn(3), d => sw(7 downto 0), load => btn(2), q => q
75             (7 downto 0));
76
77     x <= g & q;
78
79     U3 : x7segb
80         port map(
81             clk => clk190,
82             clr => btn(3),
83             x => x,
84             a_to_g => a_to_g,
85             an => an,
86             dp => dp
87         );
88
89     ld <= sw;
90
91 end Behavioral;
92
```

```
1
2  # PlanAhead Generated physical constraints
3
4  NET "a_to_g[0]" LOC = H14;
5  NET "a_to_g[1]" LOC = J17;
6  NET "a_to_g[2]" LOC = G14;
7  NET "a_to_g[3]" LOC = D16;
8  NET "a_to_g[4]" LOC = D17;
9  NET "a_to_g[5]" LOC = F18;
10 NET "a_to_g[6]" LOC = L18;
11 NET "an[0]" LOC = F17;
12 NET "an[1]" LOC = H17;
13 NET "an[2]" LOC = C18;
14 NET "an[3]" LOC = F15;
15 NET "btn[2]" LOC = E18;
16 NET "btn[3]" LOC = H13;
17 NET "dp" LOC = C17;
18 NET "ld[0]" LOC = J14;
19 NET "ld[1]" LOC = J15;
20 NET "ld[2]" LOC = K15;
21 NET "ld[3]" LOC = K14;
22 NET "ld[4]" LOC = E17;
23 NET "ld[5]" LOC = P15;
24 NET "ld[6]" LOC = F4;
25 NET "ld[7]" LOC = R4;
26 NET "mclk" LOC = B8;
27 NET "sw[0]" LOC = G18;
28 NET "sw[1]" LOC = H18;
29 NET "sw[2]" LOC = K18;
30 NET "sw[3]" LOC = K17;
31 NET "sw[4]" LOC = L14;
32 NET "sw[5]" LOC = L13;
33 NET "sw[6]" LOC = N17;
34 NET "sw[7]" LOC = R17;
35
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    16:32:58 04/08/2011
6  -- Design Name:
7  -- Module Name:    F:/Senior Project/register_top/register_top_TB.vhd
8  -- Project Name:   register_top
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: register_top
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32  ENTITY register_top_TB IS
33  END register_top_TB;
34
35  ARCHITECTURE behavior OF register_top_TB IS
36
37      -- Component Declaration for the Unit Under Test (UUT)
38
39      COMPONENT register_top
40      PORT(
41          mclk : IN  std_logic;
42          sw  : IN  std_logic_vector(7 downto 0);
43          btn : IN  std_logic_vector(3 downto 2);
44          ld  : OUT std_logic_vector(7 downto 0);
45          a_to_g : OUT std_logic_vector(6 downto 0);
46          an : OUT std_logic_vector(3 downto 0);
47          dp : OUT std_logic
48      );
49      END COMPONENT;
50
51
52      --Inputs
53      signal mclk : std_logic := '0';
54      signal sw  : std_logic_vector(7 downto 0) := (others => '0');
55      signal btn : std_logic_vector(3 downto 2) := (others => '0');
56
```

```
57  --Outputs
58  signal ld : std_logic_vector(7 downto 0);
59  signal a_to_g : std_logic_vector(6 downto 0);
60  signal an : std_logic_vector(3 downto 0);
61  signal dp : std_logic;
62
63  -- Clock period definitions
64  constant mclk_period : time := 10 ns;
65
66  BEGIN
67
68  -- Instantiate the Unit Under Test (UUT)
69  uut: register_top PORT MAP (
70      mclk => mclk,
71      sw => sw,
72      btn => btn,
73      ld => ld,
74      a_to_g => a_to_g,
75      an => an,
76      dp => dp
77  );
78
79  -- Clock process definitions
80  mclk_process :process
81  begin
82      mclk <= '0';
83      wait for mclk_period/2;
84      mclk <= '1';
85      wait for mclk_period/2;
86  end process;
87
88
89  -- Stimulus process
90  stim_proc: process
91  begin
92      btn(3) <= '0';
93
94      sw <="00000000";
95
96      for i in 0 to 16 loop
97          wait for 10 ns;
98          sw <= std_logic_vector(unsigned(sw) +1);
99      end loop;
100
101      wait;
102  end process;
103
104  END;
105
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    00:23:27 04/09/2011
6  -- Design Name:
7  -- Module Name:    F:/Senior Project/shift4/shift4_TB.vhd
8  -- Project Name:   shift4
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: shift4
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30
31  -- Uncomment the following library declaration if using
32  -- arithmetic functions with Signed or Unsigned values
33  --USE ieee.numeric_std.ALL;
34
35  ENTITY shift4_TB IS
36  END shift4_TB;
37
38  ARCHITECTURE behavior OF shift4_TB IS
39
40      -- Component Declaration for the Unit Under Test (UUT)
41
42      COMPONENT shift4
43      PORT(
44          clk : IN  std_logic;
45          clr : IN  std_logic;
46          data_in : IN  std_logic;
47          q : OUT  std_logic_vector(3 downto 0)
48      );
49      END COMPONENT;
50
51
52      --Inputs
53      signal clk : std_logic := '0';
54      signal clr : std_logic := '0';
55      signal data_in : std_logic := '0';
56
```



```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    00:13:43 04/09/2011
6  -- Design Name:
7  -- Module Name:    shift4 - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity shift4 is
24     Port ( clk : in  STD_LOGIC;
25           clr : in  STD_LOGIC;
26           data_in : in  STD_LOGIC;
27           q : out  STD_LOGIC_VECTOR (3 downto 0));
28 end shift4;
29
30 architecture Behavioral of shift4 is
31
32     signal qs : STD_LOGIC_VECTOR(3 downto 0);
33
34     begin
35         --4-bit shift register
36         process(clk, clr)
37         begin
38             if clr = '1' then
39                 qs <= "0000";
40             elsif clk'event and clk = '1' then
41                 qs(3) <= data_in;
42                 qs(2 downto 0) <= qs(3 downto 1);
43             end if;
44         end process;
45         q <= qs;
46
47     end Behavioral;
48
49
```

```
57      --Outputs
58      signal q : std_logic_vector(3 downto 0);
59
60      -- Clock period definitions
61      constant clk_period : time := 10 ns;
62
63      BEGIN
64
65      -- Instantiate the Unit Under Test (UUT)
66      uut: shift4 PORT MAP (
67          clk => clk,
68          clr => clr,
69          data_in => data_in,
70          q => q
71      );
72
73      -- Clock process definitions
74      clk_process :process
75      begin
76          clk <= '0';
77          wait for clk_period/2;
78          clk <= '1';
79          wait for clk_period/2;
80      end process;
81
82      data_process: process
83      begin
84          data_in <= '1';
85          wait for 25 ns;
86          data_in <= '0';
87          wait for 75 ns;
88          data_in <= '1';
89          wait for 20 ns;
90          data_in <= '0';
91          wait for 20 ns;
92          data_in <= '1';
93          wait for 20 ns;
94          data_in <= '0';
95          wait for 20 ns;
96          data_in <= '1';
97          wait for 20 ns;
98          data_in <= '0';
99          wait for 100 ns;
100
101      end process;
102
103
104
105      -- Stimulus process
106      stim_proc: process
107      begin
108          clr <= '1';
109          wait for 10 ns;
110          clr <= '0';
111          wait for 100 ns;
112          wait;
```

```
113     end process;  
114  
115     END;  
116
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    00:29:16 04/09/2011
6  -- Design Name:
7  -- Module Name:    shiftReg - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23
24 entity shiftReg is
25     generic(N:integer :=8);
26     Port ( clk : in  STD_LOGIC;
27           clr : in  STD_LOGIC;
28           data_in : in  STD_LOGIC;
29           q : out  STD_LOGIC_VECTOR (N-1 downto 0));
30 end shiftReg;
31
32 architecture Behavioral of shiftReg is
33     signal qs : STD_LOGIC_VECTOR(N-1 downto 0);
34     begin
35         --N-bit shift register
36         process(clk, clr)
37         begin
38             if clr = '1' then
39                 qs <= (others =>'0');
40             elsif clk'event and clk = '1' then
41                 qs(N-1) <= data_in;
42                 qs(N-2 downto 0) <= qs(N-1 downto 1);
43             end if;
44         end process;
45
46         q <= qs;
47
48
49     end Behavioral;
50
51
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    21:42:27 04/17/2011
6  -- Design Name:
7  -- Module Name:    ring4 - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity ring4 is
24     Port ( clk : in  STD_LOGIC;
25           clr : in  STD_LOGIC;
26           q : out  STD_LOGIC_VECTOR (3 downto 0));
27 end ring4;
28
29 architecture Behavioral of ring4 is
30
31     signal qs : STD_LOGIC_VECTOR(3 downto 0);
32
33     begin
34
35         --4-bit ring counter
36         process(clk, clr)
37         begin
38             if clr = '1' then
39                 qs <= "0000";
40                 qs(0) <= '1';
41             elsif clk'event and clk = '1' then
42                 qs(3) <= qs(0);
43                 qs(2 downto 0) <= qs(3 downto 1);
44             end if;
45         end process;
46
47         q <= qs;
48     end Behavioral;
49
50
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    22:01:33 04/17/2011
6  -- Design Name:
7  -- Module Name:    F:/Senior Project/Example 22/ring4/ring4_TB.vhd
8  -- Project Name:   ring4
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: ring4
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30
31  -- Uncomment the following library declaration if using
32  -- arithmetic functions with Signed or Unsigned values
33  --USE ieee.numeric_std.ALL;
34
35  ENTITY ring4_TB IS
36  END ring4_TB;
37
38  ARCHITECTURE behavior OF ring4_TB IS
39
40      -- Component Declaration for the Unit Under Test (UUT)
41
42      COMPONENT ring4
43      PORT(
44          clk : IN  std_logic;
45          clr : IN  std_logic;
46          q  : OUT  std_logic_vector(3 downto 0)
47      );
48      END COMPONENT;
49
50
51      --Inputs
52      signal clk : std_logic := '0';
53      signal clr : std_logic := '0';
54
55      --Outputs
56      signal q : std_logic_vector(3 downto 0);
```

```
57
58     -- Clock period definitions
59     constant clk_period : time := 10 ns;
60
61 BEGIN
62
63     -- Instantiate the Unit Under Test (UUT)
64     uut: ring4 PORT MAP (
65         clk => clk,
66         clr => clr,
67         q => q
68     );
69
70     -- Clock process definitions
71     clk_process :process
72     begin
73         clk <= '0';
74         wait for clk_period/2;
75         clk <= '1';
76         wait for clk_period/2;
77     end process;
78
79
80     -- Stimulus process
81     stim_proc: process
82     begin
83         clr <= '1';
84         wait for 10 ns;
85         clr <= '0';
86         wait for 100 ns;
87
88         wait;
89     end process;
90
91 END;
92
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    00:34:34 04/09/2011
6  -- Design Name:
7  -- Module Name:    ring - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity ring is
24     generic( N:integer := 8);
25     Port ( clk : in  STD_LOGIC;
26           clr : in  STD_LOGIC;
27           q : out  STD_LOGIC_VECTOR (N-1 downto 0));
28 end ring;
29
30 architecture Behavioral of ring is
31
32     signal qs : STD_LOGIC_VECTOR(N-1 downto 0);
33
34     begin
35         --N-bit shift register
36         process(clk, clr)
37         begin
38             if clr = '1' then
39                 qs <= (others => '0');
40                 qs(0) <= '1';
41             elsif clk'event and clk = '1' then
42                 qs(N-1) <= qs(0);
43                 qs(N-2 downto 0) <= qs(N-1 downto 1);
44             end if;
45         end process;
46
47         q <= qs;
48     end Behavioral;
49
50
```



```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    00:37:50 04/09/2011
6  -- Design Name:
7  -- Module Name:    F:/Senior Project/ring/ring_TB.vhd
8  -- Project Name:   ring
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: ring
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32  ENTITY ring_TB IS
33  END ring_TB;
34
35  ARCHITECTURE behavior OF ring_TB IS
36
37      -- Component Declaration for the Unit Under Test (UUT)
38
39      COMPONENT ring
40      PORT(
41          clk : IN  std_logic;
42          clr : IN  std_logic;
43          q : OUT  std_logic_vector(7 downto 0)
44      );
45      END COMPONENT;
46
47
48      --Inputs
49      signal clk : std_logic := '0';
50      signal clr : std_logic := '0';
51
52      --Outputs
53      signal q : std_logic_vector(7 downto 0);
54
55      -- Clock period definitions
56      constant clk_period : time := 10 ns;
```

```
57
58 BEGIN
59
60     -- Instantiate the Unit Under Test (UUT)
61     uut: ring PORT MAP (
62         clk => clk,
63         clr => clr,
64         q => q
65     );
66
67     -- Clock process definitions
68     clk_process :process
69     begin
70         clk <= '0';
71         wait for clk_period/2;
72         clk <= '1';
73         wait for clk_period/2;
74     end process;
75
76     -- Stimulus process
77     stim_proc: process
78     begin
79         clr <= '1';
80         wait for 10 ns;
81         clr <= '0';
82         wait for 100 ns;
83
84         wait;
85     end process;
86
87
88 END;
89
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    00:44:05 04/09/2011
6  -- Design Name:
7  -- Module Name:    ring8_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity ring8_top is
24     Port ( mclk : in  STD_LOGIC;
25           btn : in  STD_LOGIC_VECTOR (3 downto 3);
26           ld : out STD_LOGIC_VECTOR (7 downto 0));
27 end ring8_top;
28
29 architecture Behavioral of ring8_top is
30 component ring is
31     generic(N:integer := 8);
32     port(
33         clk : in STD_LOGIC;
34         clr : in  STD_LOGIC;
35         q : out STD_LOGIC_VECTOR(N-1 downto 0)
36     );
37 end component;
38
39 component clkdiv is
40     port(
41         mclk : in STD_LOGIC;
42         clr : in STD_LOGIC;
43         clk3 : out STD_LOGIC
44     );
45 end component;
46
47 signal clk3 : STD_LOGIC;
48
49 begin
50
51 U1 : ring
52     generic map ( N => 8)
53     port map ( clk => clk3, clr => btn(3), q => ld(7 downto 0));
54
55 U2 : clkdiv
56     port map( mclk => mclk, clr => btn(3), clk3 => clk3);
```

57

58 end Behavioral;

59

60

```
1
2 # PlanAhead Generated physical constraints
3
4 NET "mclk" LOC = B8;
5 NET "btn[3]" LOC = H13;
6 NET "ld[0]" LOC = J14;
7 NET "ld[1]" LOC = J15;
8 NET "ld[2]" LOC = K15;
9 NET "ld[3]" LOC = K14;
10 NET "ld[4]" LOC = E17;
11 NET "ld[5]" LOC = P15;
12 NET "ld[6]" LOC = F4;
13 NET "ld[7]" LOC = R4;
14
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    00:44:05 04/09/2011
6  -- Design Name:
7  -- Module Name:    ring8_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity ring8_top is
24     Port ( mclk : in  STD_LOGIC;
25           btn : in  STD_LOGIC_VECTOR (3 downto 3);
26           ld : out STD_LOGIC_VECTOR (7 downto 0));
27 end ring8_top;
28
29 architecture Behavioral of ring8_top is
30 component ring is
31     generic(N:integer := 8);
32     port(
33         clk : in STD_LOGIC;
34         clr : in  STD_LOGIC;
35         q : out STD_LOGIC_VECTOR(N-1 downto 0)
36     );
37 end component;
38
39 component clkdiv is
40     port(
41         mclk : in STD_LOGIC;
42         clr : in STD_LOGIC;
43         clk3 : out STD_LOGIC
44     );
45 end component;
46
47 signal clk3 : STD_LOGIC;
48
49 begin
50
51 U1 : ring
52     generic map ( N => 8)
53     port map ( clk => clk3, clr => btn(3), q => ld(7 downto 0));
54
55 U2 : clkdiv
56     port map( mclk => mclk, clr => btn(3), clk3 => clk3);
```

57

58 end Behavioral;

59

60

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    01:08:46 04/09/2011
6  -- Design Name:
7  -- Module Name:    johnson - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity johnson is
24     generic( N:integer := 8);
25     Port ( clk : in  STD_LOGIC;
26           clr : in  STD_LOGIC;
27           q : out STD_LOGIC_VECTOR (N-1 downto 0));
28 end johnson;
29
30 architecture Behavioral of johnson is
31     signal qs : STD_LOGIC_VECTOR(N-1 downto 0);
32     begin
33
34     --N-bit Johnson counter
35     process(clk, clr)
36     begin
37         if clr = '1' then
38             qs <= (others => '0');
39         elsif clk'event and clk = '1' then
40             qs(N-1) <= not qs(0);
41             qs(N-2 downto 0) <= qs(N-1 downto 1);
42         end if;
43     end process;
44     q <= qs;
45 end Behavioral;
46
47
```



```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    12:23:35 04/18/2011
6  -- Design Name:
7  -- Module Name:    E:/Senior Project/Example 23/johnson/johnson_TB.vhd
8  -- Project Name:   johnson
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: johnson
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30
31  ENTITY johnson_TB IS
32  END johnson_TB;
33
34  ARCHITECTURE behavior OF johnson_TB IS
35
36      -- Component Declaration for the Unit Under Test (UUT)
37
38      COMPONENT johnson
39      PORT(
40          clk : IN  std_logic;
41          clr : IN  std_logic;
42          q : OUT  std_logic_vector(7 downto 0)
43      );
44      END COMPONENT;
45
46
47      --Inputs
48      signal clk : std_logic := '0';
49      signal clr : std_logic := '0';
50
51      --Outputs
52      signal q : std_logic_vector(7 downto 0);
53
54      -- Clock period definitions
55      constant clk_period : time := 10 ns;
56
```

```
57 BEGIN
58
59     -- Instantiate the Unit Under Test (UUT)
60     uut: johnson PORT MAP (
61         clk => clk,
62         clr => clr,
63         q => q
64     );
65
66     -- Clock process definitions
67     clk_process :process
68     begin
69         clk <= '0';
70         wait for clk_period/2;
71         clk <= '1';
72         wait for clk_period/2;
73     end process;
74
75     -- Stimulus process
76     stim_proc: process
77     begin
78         clr <= '1';
79         wait for 10 ns;
80         clr <= '0';
81         wait for 100 ns;
82         wait;
83     end process;
84
85
86 END;
87
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    01:15:18 04/09/2011
6  -- Design Name:
7  -- Module Name:    johnson8_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity johnson8_top is
24     Port ( mclk : in  STD_LOGIC;
25           btn  : in  STD_LOGIC_VECTOR (3 downto 3);
26           ld   : out STD_LOGIC_VECTOR (7 downto 0));
27 end johnson8_top;
28
29 architecture Behavioral of johnson8_top is
30
31     component johnson is
32         generic(N:integer := 8);
33         port(
34             clk : in STD_LOGIC;
35             clr : in STD_LOGIC;
36             q   : out STD_LOGIC_VECTOR(N-1 downto 0)
37         );
38     end component;
39
40     component clkdiv is
41         port(
42             mclk : in STD_LOGIC;
43             clr  : in STD_LOGIC;
44             clk3 : out STD_LOGIC
45         );
46     end component;
47
48     signal clk3 : STD_LOGIC;
49
50     begin
51
52     U1 : johnson
53         generic map ( N => 8)
54         port map ( clk => clk3, clr => btn(3), q => ld(7 downto 0));
55
56     U2 : clkdiv
```

```
57     port map( mclk => mclk, clr => btn(3), clk3 => clk3);  
58  
59 end Behavioral;  
60  
61
```

```
1  NET "mclk" LOC = B8;  
2  NET "btn{3}" LOC = H13;  
3  NET "ld[0]" LOC = J14;  
4  NET "ld[1]" LOC = J15;  
5  NET "ld[2]" LOC = K15;  
6  NET "ld[3]" LOC = K14;  
7  NET "ld[4]" LOC = E17;  
8  NET "ld[5]" LOC = P15;  
9  NET "ld[6]" LOC = F4;  
10 NET "ld[7]" LOC = R4;  
11
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    01:25:43 04/09/2011
6  -- Design Name:
7  -- Module Name:    debounce4 - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity debounce4 is
24     Port ( inp : in  STD_LOGIC_VECTOR (3 downto 0);
25           cclk : in  STD_LOGIC;
26           clr  : in  STD_LOGIC;
27           outp : out STD_LOGIC_VECTOR (3 downto 0));
28 end debounce4;
29
30 architecture Behavioral of debounce4 is
31     signal delay1, delay2, delay3 : STD_LOGIC_VECTOR(3 downto 0);
32 begin
33     process(cclk, clr)
34     begin
35         if clr = '1' then
36             delay1 <= "0000";
37             delay2 <= "0000";
38             delay3 <= "0000";
39         elsif cclk'event and cclk = '1' then
40             delay1 <= inp;
41             delay2 <= delay1;
42             delay3 <= delay2;
43         end if;
44     end process;
45
46     outp <= delay1 and delay2 and delay3;
47
48 end Behavioral;
49
50
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    12:57:59 04/18/2011
6  -- Design Name:
7  -- Module Name:    E:/Senior Project/Example 24/debounce4/debouncer4_TB.vhd
8  -- Project Name:   debounce4
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: debounce4
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30
31  ENTITY debouncer4_TB IS
32  END debouncer4_TB;
33
34  ARCHITECTURE behavior OF debouncer4_TB IS
35
36      -- Component Declaration for the Unit Under Test (UUT)
37
38      COMPONENT debounce4
39      PORT(
40          inp : IN  std_logic_vector(3 downto 0);
41          cclk : IN  std_logic;
42          clr  : IN  std_logic;
43          outp : OUT std_logic_vector(3 downto 0)
44      );
45      END COMPONENT;
46
47
48      --Inputs
49      signal cclk : std_logic := '0';
50      signal clr  : std_logic := '0';
51      signal inp  : std_logic_vector(3 downto 0) := (others => '0');
52
53      --Outputs
54      signal outp : std_logic_vector(3 downto 0);
55
56      -- Clock period definitions
```

```
57     constant cclk_period : time := 10 ns;
58
59 BEGIN
60
61     -- Instantiate the Unit Under Test (UUT)
62     uut: debounce4 PORT MAP (
63         inp => inp,
64         cclk => cclk,
65         clr => clr,
66         outp => outp
67     );
68
69     -- Clock process definitions
70     cclk_process :process
71     begin
72         cclk <= '0';
73         wait for cclk_period/2;
74         cclk <= '1';
75         wait for cclk_period/2;
76     end process;
77
78     inp_process :process
79     begin
80         inp <= "0000";
81         wait for 125 ns;
82         inp <= "0001";
83         wait for 25 ns;
84         inp <= "0000";
85         wait for 25 ns;
86         inp <= "0001";
87         wait for 10 ns;
88         inp <= "0001";
89         wait for 150 ns;
90         inp <= "0000";
91         wait for 25 ns;
92         inp <= "0000";
93         wait for 100 ns;
94     end process;
95
96     -- Stimulus process
97     stim_proc: process
98     begin
99         clr <= '1';
100         wait for 10 ns;
101         clr <= '0';
102         wait for 100 ns;
103
104         wait;
105     end process;
106
107 END;
```



```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    01:31:36 04/09/2011
6  -- Design Name:
7  -- Module Name:    clock_pulse - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 Entity clock_pulse is
24     Port ( inp : in  STD_LOGIC;
25           cclk : in  STD_LOGIC;
26           clr  : in  STD_LOGIC;
27           outp : out STD_LOGIC);
28 end clock_pulse;
29
30 architecture Behavioral of clock_pulse is
31     signal delay1, delay2, delay3 : STD_LOGIC;
32
33     begin
34         process(cclk, clr)
35         begin
36             if clr = '1' then
37                 delay1 <= '0';
38                 delay2 <= '0';
39                 delay3 <= '0';
40             elsif cclk'event and cclk = '1' then
41                 delay1 <= inp;
42                 delay2 <= delay1;
43                 delay3 <= delay2;
44             end if;
45         end process;
46
47         outp <= delay1 and delay2 and not delay3;
48
49     end Behavioral;
50
51
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    01:35:53 04/09/2011
6  -- Design Name:
7  -- Module Name:    F:/Senior Project/clock_pulse/clock_puse_TB.vhd
8  -- Project Name:   clock_pulse
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: clock_pulse
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30
31  -- Uncomment the following library declaration if using
32  -- arithmetic functions with Signed or Unsigned values
33  --USE ieee.numeric_std.ALL;
34
35  ENTITY clock_puse_TB IS
36  END clock_puse_TB;
37
38  ARCHITECTURE behavior OF clock_puse_TB IS
39
40      -- Component Declaration for the Unit Under Test (UUT)
41
42      COMPONENT clock_pulse
43      PORT(
44          inp : IN  std_logic;
45          cclk : IN  std_logic;
46          clr : IN  std_logic;
47          outp : OUT std_logic
48      );
49      END COMPONENT;
50
51
52      --Inputs
53      signal cclk : std_logic := '0';
54      signal clr : std_logic := '0';
55      signal inp : std_logic := '0';
56
```

```
57  --Outputs
58  signal outp : std_logic;
59
60  -- Clock period definitions
61  constant cclk_period : time := 10 ns;
62
63  BEGIN
64
65  -- Instantiate the Unit Under Test (UUT)
66  uut: clock_pulse PORT MAP (
67      inp => inp,
68      cclk => cclk,
69      clr => clr,
70      outp => outp
71  );
72
73  -- Clock process definitions
74  cclk_process :process
75  begin
76      cclk <= '0';
77      wait for cclk_period/2;
78      cclk <= '1';
79      wait for cclk_period/2;
80  end process;
81
82
83  -- Stimulus process
84  stim_proc: process
85  begin
86      clr <= '1';
87      wait for 10 ns;
88      clr <= '0';
89
90
91      inp <= '0';
92      wait for 100 ns;
93      inp <= '1';
94      wait for 200 ns;
95      inp <= '0';
96      wait;
97  end process;
98
99  END;
100
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    01:43:00 04/09/2011
6  -- Design Name:
7  -- Module Name:    clock_pulse_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity clock_pulse_top is
24     Port ( mclk : in  STD_LOGIC;
25           btn : in  STD_LOGIC_VECTOR (3 downto 2);
26           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
27           an : out  STD_LOGIC_VECTOR (3 downto 0);
28           dp : out  STD_LOGIC);
29 end clock_pulse_top;
30
31 architecture Behavioral of clock_pulse_top is
32 component clkdiv is
33     Port ( mclk : in  STD_LOGIC;
34           clr : in  STD_LOGIC;
35           clk190 : out STD_LOGIC
36           );
37 end component;
38
39 component clock_pulse is
40     Port ( inp : in  STD_LOGIC;
41           cclk : in  STD_LOGIC;
42           clr : in  STD_LOGIC;
43           outp : out STD_LOGIC);
44 end component;
45
46 component counter is
47     generic(N : integer := 16);
48     Port (
49         clr : in  STD_LOGIC;
50         clk : in  STD_LOGIC;
51         q_counter : out STD_LOGIC_VECTOR(N-1 downto 0)
52     );
53 end component;
54
55 component x7segb is
56     Port ( x : in  STD_LOGIC_VECTOR (15 downto 0);
```

```
57         clk : in  STD_LOGIC;
58         clr : in  STD_LOGIC;
59         a_to_g : out  STD_LOGIC_VECTOR (6 downto 0);
60         an : out  STD_LOGIC_VECTOR (3 downto 0);
61         dp : out  STD_LOGIC);
62     end component;
63
64     signal clk190, clkp : STD_LOGIC;
65     signal x : STD_LOGIC_VECTOR(15 downto 0);
66
67     begin
68
69     U1 : clock_pulse
70         port map( cclk => clk190, clr => btn(3), inp => btn(2), outp => clkp);
71     U2 : clkdiv
72         port map( mclk => mclk, clr => btn(3), clk190 => clk190);
73     U3 : counter
74         generic map( N => 16)
75         port map( clk => clkp, clr => btn(3), q_counter => x(15 downto 0));
76     U4 : x7segb
77         port map( clk => clk190, clr => btn(3), x => x, a_to_g => a_to_g, an => an, dp =>
dp);
78
79     end Behavioral;
80
81
```

```
1
2  # PlanAhead Generated physical constraints
3
4  NET "a_to_g{0}" LOC = H14;
5  NET "a_to_g{1}" LOC = J17;
6  NET "a_to_g{2}" LOC = G14;
7  NET "a_to_g{3}" LOC = D16;
8  NET "a_to_g{4}" LOC = D17;
9  NET "a_to_g{5}" LOC = F18;
10 NET "a_to_g{6}" LOC = L18;
11 NET "an{0}" LOC = F17;
12 NET "an{1}" LOC = H17;
13 NET "an{2}" LOC = C18;
14 NET "an{3}" LOC = F15;
15 NET "btn{2}" LOC = E18;
16 NET "btn{3}" LOC = H13;
17 NET "dp" LOC = C17;
18 NET "mclk" LOC = B8;
19
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    02:17:10 04/09/2011
6  -- Design Name:
7  -- Module Name:    eat - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity eat is
24     Port ( q : in  STD_LOGIC_VECTOR (3 downto 0);
25           eat : out STD_LOGIC);
26 end eat;
27
28 architecture Behavioral of eat is
29
30 begin
31     eat <= (not q(3) and not q(2) and q(1) and q(0))
32           or (q(2) and q(1) and not q(0))
33           or (q(3) and q(2) and q(0))
34           or (q(3) and not q(1) and q(0))
35           or (q(3) and not q(2) and not q(0));
36
37 end Behavioral;
38
39
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    02:25:40 04/09/2011
6  -- Design Name:
7  -- Module Name:    eat_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity eat_top is
24     Port ( mclk : in  STD_LOGIC;
25           btn : in  STD_LOGIC_VECTOR (3 downto 3);
26           ld : out  STD_LOGIC_VECTOR (0 downto 0));
27 end eat_top;
28
29 architecture Behavioral of eat_top is
30 component clkdiv is
31     Port ( mclk : in  STD_LOGIC;
32           clr : in  STD_LOGIC;
33           clk3 : out  STD_LOGIC
34           );
35 end component;
36
37 component counter is
38     generic(N : integer := 4);
39     Port (
40         clr : in  STD_LOGIC;
41         clk : in  STD_LOGIC;
42         q_counter : out  STD_LOGIC_VECTOR(N-1 downto 0)
43         );
44 end component;
45
46 component eat is
47     port(
48         q : in STD_LOGIC_VECTOR(3 downto 0);
49         eat : out STD_LOGIC
50     );
51 end component;
52
53 signal clk3, clr : STD_LOGIC;
54 signal q : STD_LOGIC_VECTOR(3 downto 0);
55
56 begin
```



```
57
58   clr <= btn(3);
59
60   U1 : eat
61       port map( q => q, eat => ld(0));
62   U2 : clkdiv
63       port map( mclk => mclk, clr => clr, clk3 => clk3);
64   U3 : counter
65       generic map( N => 4)
66       port map( clk => clk3, clr => clr, q_counter => q(3 downto 0));
67
68   end Behavioral;
69
70
```

```
1
2
3  NET "mclk" LOC = B8;
4  NET "btn[3]" LOC = H13;
5  NET "ld[0]" LOC = J14;
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    02:41:56 04/09/2011
6  -- Design Name:
7  -- Module Name:    shift_array - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity shift_array is
24     Port ( clk : in  STD_LOGIC;
25           clr : in  STD_LOGIC;
26           x : out  STD_LOGIC_VECTOR (15 downto 0));
27 end shift_array;
28
29 architecture Behavioral of shift_array is
30     signal msg_array : STD_LOGIC_VECTOR( 0 TO 63);
31     constant PHONE_NO : STD_LOGIC_VECTOR(63 downto 0) :=X"555DS67D5309FFFF";
32
33     begin
34         process(clk, clr)
35         begin
36             if clr = '1' then
37                 msg_array <= PHONE_NO;
38             elsif(clk'event and clk = '1') then
39                 msg_array(0 to 59) <= msg_array(4 to 63);
40                 msg_array(60 to 63) <= msg_array(0 to 3);
41             end if;
42         end process;
43         x <= msg_array(0 to 15);
44
45
46     end Behavioral;
47
48
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   15:38:34 04/18/2011
6  -- Design Name:
7  -- Module Name:   H:/Senior Project/Example 29/shift_array/shift_array_TB.vhd
8  -- Project Name:  shift_array
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: shift_array
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30
31  -- Uncomment the following library declaration if using
32  -- arithmetic functions with Signed or Unsigned values
33  --USE ieee.numeric_std.ALL;
34
35  ENTITY shift_array_TB IS
36  END shift_array_TB;
37
38  ARCHITECTURE behavior OF shift_array_TB IS
39
40      -- Component Declaration for the Unit Under Test (UUT)
41
42      COMPONENT shift_array
43      PORT(
44          clk : IN  std_logic;
45          clr : IN  std_logic;
46          x : OUT  std_logic_vector(15 downto 0)
47      );
48      END COMPONENT;
49
50
51      --Inputs
52      signal clk : std_logic := '0';
53      signal clr : std_logic := '0';
54
55      --Outputs
56      signal x : std_logic_vector(15 downto 0);
```

```
57
58     -- Clock period definitions
59     constant clk_period : time := 10 ns;
60
61 BEGIN
62
63     -- Instantiate the Unit Under Test (UUT)
64     uut: shift_array PORT MAP (
65         clk => clk,
66         clr => clr,
67         x => x
68     );
69
70     -- Clock process definitions
71     clk_process :process
72     begin
73         clk <= '0';
74         wait for clk_period/2;
75         clk <= '1';
76         wait for clk_period/2;
77     end process;
78
79
80     -- Stimulus process
81     stim_proc: process
82     begin
83         clr <= '1';
84         wait for 10 ns;
85         clr <= '0';
86         wait for 100 ns;
87
88         wait;
89     end process;
90
91 END;
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    02:46:40 04/09/2011
6  -- Design Name:
7  -- Module Name:    x7seg_msg - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22 use IEEE.STD_LOGIC_UNSIGNED.ALL;
23
24 entity x7seg_msg is
25     Port ( x : in  STD_LOGIC_VECTOR (15 downto 0);
26           clk : in  STD_LOGIC;
27           clr : in  STD_LOGIC;
28           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
29           an : out  STD_LOGIC_VECTOR (3 downto 0);
30           dp : out  STD_LOGIC);
31 end x7seg_msg;
32
33 architecture Behavioral of x7seg_msg is
34     signal s : STD_LOGIC_VECTOR(1 downto 0);
35     signal digit , aen : STD_LOGIC_VECTOR(3 downto 0);
36
37     begin
38
39     aen <= "1111";
40     dp <= '1';
41
42     --Quad 4-to-1 MUX: mux44
43     process(s, x)
44     begin
45         case(s) is
46             when "00" => digit <= x(3 downto 0);
47             when "01" => digit <= x(7 downto 4);
48             when "10" => digit <= x(11 downto 8);
49             when others => digit <= x(15 downto 12);
50         end case;
51     end process;
52
53     --2-bit counter
54     process(clk, clr)
55     begin
56         if clr = '1' then
```

```
57         s <= "00";
58     elsif clk'event and clk = '1' then
59         s <= s + 1;
60     end if;
61 end process;
62
63 --Digit select : ancode
64 process(s, aen)
65 begin
66     an <= "1111";
67     if aen(conv_integer(s)) = '1' then
68         an(conv_integer(s)) <= '0';
69     end if;
70 end process;
71
72 --7-segment decoder : hex7seg
73 process(digit)
74 begin
75     case digit is
76         when X"0" => a_to_g <= "0000001"; --0
77         when X"1" => a_to_g <= "1001111"; --1
78         when X"2" => a_to_g <= "0010010"; --2
79         when X"3" => a_to_g <= "0000110"; --3
80         when X"4" => a_to_g <= "1001100"; --4
81         when X"5" => a_to_g <= "0100100"; --5
82         when X"6" => a_to_g <= "0100000"; --6
83         when X"7" => a_to_g <= "0001101"; --7
84         when X"8" => a_to_g <= "0000000"; --8
85         when X"9" => a_to_g <= "0000100"; --9
86         when X"A" => a_to_g <= "0001000"; --A
87         when X"B" => a_to_g <= "1100000"; --B
88         when X"C" => a_to_g <= "0110001"; --C
89         when X"D" => a_to_g <= "1111110"; --dash
90         when X"E" => a_to_g <= "1111111"; --E
91         when others => a_to_g <= "1111111"; --F
92     end case;
93 end process;
94
95 end Behavioral;
96
97
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    02:55:54 04/09/2011
6  -- Design Name:
7  -- Module Name:    scoll_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity scoll_top is
24     Port ( mclk : in  STD_LOGIC;
25           btn : in  STD_LOGIC_VECTOR (3 downto 3);
26           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
27           an : out  STD_LOGIC_VECTOR (3 downto 0);
28           dp : out  STD_LOGIC);
29 end scoll_top;
30
31 architecture Behavioral of scoll_top is
32 component clkdiv is
33     port(
34         mclk : in STD_LOGIC;
35         clr : in STD_LOGIC;
36         clk190 : out STD_LOGIC;
37         clk3 : out STD_LOGIC
38     );
39 end component;
40
41 component shift_array is
42     port(
43         clk : STD_LOGIC;
44         clr : in STD_LOGIC;
45         x : out STD_LOGIC_VECTOR(15 downto 0)
46     );
47 end component;
48
49 component x7seg_msg is
50     port(
51         x : in  STD_LOGIC_VECTOR (15 downto 0);
52         clk : in  STD_LOGIC;
53         clr : in  STD_LOGIC;
54         a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
55         an : out  STD_LOGIC_VECTOR (3 downto 0);
56         dp : out  STD_LOGIC);
```



```
57  end component;
58
59  signal clr, clk190, clk3 : STD_LOGIC;
60  signal x : STD_LOGIC_VECTOR(15 downto 0);
61
62  begin
63
64      clr <= btn(3);
65
66      U1 : clkdiv
67      port map(
68          mclk => mclk, clr => clr, clk190 => clk190, clk3 => clk3
69      );
70      U2 : shift_array
71      port map(
72          clk => clk3, clr => clr, x => x
73      );
74      U3 : x7seg_msg
75      port map(
76          x => x, clk => clk190, clr => clr, a_to_g => a_to_g, an => an, dp => dp
77      );
78
79  end Behavioral;
80
81
```

```
1  NET "a_to_g[0]" LOC = H14;
2  NET "a_to_g[1]" LOC = J17;
3  NET "a_to_g[2]" LOC = G14;
4  NET "a_to_g[3]" LOC = D16;
5  NET "a_to_g[4]" LOC = D17;
6  NET "a_to_g[5]" LOC = F18;
7  NET "a_to_g[6]" LOC = L18;
8  NET "an[0]" LOC = F17;
9  NET "an[1]" LOC = H17;
10 NET "an[2]" LOC = C18;
11 NET "an[3]" LOC = F15;
12 NET "btn[3]" LOC = H13;
13 NET "dp" LOC = C17;
14 NET "mclk" LOC = B8;
15
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    16:15:36 04/10/2011
6  -- Design Name:
7  -- Module Name:    fib - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22 use IEEE.STD_LOGIC_UNSIGNED.ALL;
23
24 entity fib is
25     Port ( clk : in  STD_LOGIC;
26           clr : in  STD_LOGIC;
27           f : out  STD_LOGIC_VECTOR (15 downto 0));
28 end fib;
29
30 architecture Behavioral of fib is
31     signal fn1, fn : STD_LOGIC_VECTOR(15 downto 0);
32     constant fib_max: integer := 9999;
33
34     begin
35     P1: process(clk, clr)
36     begin
37         if clr = '1' then
38             fn <= "0000000000000000";
39             fn1 <= "0000000000000001";
40         elsif clk'event and clk = '1' then
41             if fn1 < fib_max then
42                 fn1 <= fn1 + fn;
43                 fn <= fn1;
44             else
45                 fn <= "0000000000000000";
46                 fn1 <= "0000000000000001";
47             end if;
48         end if;
49     end process;
50
51     f <= fn;
52
53 end Behavioral;
54
55
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    16:27:45 04/10/2011
6  -- Design Name:
7  -- Module Name:    fib_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity fib_top is
24     Port ( mclk : in  STD_LOGIC;
25           btn : in  STD_LOGIC_VECTOR (3 downto 3);
26           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
27           an : out  STD_LOGIC_VECTOR (3 downto 0);
28           dp : out  STD_LOGIC);
29 end fib_top;
30
31 architecture Behavioral of fib_top is
32 component x7segb is
33     Port ( x : in  STD_LOGIC_VECTOR (15 downto 0);
34           clk : in  STD_LOGIC;
35           clr : in  STD_LOGIC;
36           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
37           an : out  STD_LOGIC_VECTOR (3 downto 0);
38           dp : out  STD_LOGIC
39     );
40 end component;
41
42 component clkdiv is
43     Port ( mclk : in  STD_LOGIC;
44           clr : in  STD_LOGIC;
45           clk190 : out STD_LOGIC;
46           clk3 : out STD_LOGIC
47     );
48 end component;
49
50 component fib is
51     Port ( clk : in  STD_LOGIC;
52           clr : in  STD_LOGIC;
53           f : out  STD_LOGIC_VECTOR (15 downto 0)
54     );
55 end component;
56
```

```
57 signal x : STD_LOGIC_VECTOR (15 downto 0);
58 signal clk190, clk3, clr : STD_LOGIC;
59
60 begin
61   clr <= btn(3);
62
63   U1: x7segb
64     port map(
65       clk => clk190,
66       clr => clr,
67       x => x,
68       a_to_g => a_to_g,
69       an => an,
70       dp => dp
71     );
72
73   U2: clkdiv
74     port map(
75       mclk => mclk,
76       clr => clr,
77       clk190 => clk190,
78       clk3 => clk3
79     );
80
81   U3: fib
82     port map(
83       clk => clk3,
84       clr => clr,
85       f => x
86     );
87
88 end Behavioral;
89
```

```
1  NET "a_to_g[0]" LOC = H14;
2  NET "a_to_g[1]" LOC = J17;
3  NET "a_to_g[2]" LOC = G14;
4  NET "a_to_g[3]" LOC = D16;
5  NET "a_to_g[4]" LOC = D17;
6  NET "a_to_g[5]" LOC = F18;
7  NET "a_to_g[6]" LOC = L18;
8  NET "an[0]" LOC = F17;
9  NET "an[1]" LOC = H17;
10 NET "an[2]" LOC = C18;
11 NET "an[3]" LOC = F15;
12 NET "btn[3]" LOC = H13;
13 NET "dp" LOC = C17;
14 NET "mclk" LOC = B8;
15
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    21:32:43 03/22/2011
6  -- Design Name:
7  -- Module Name:    clkdiv - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22 use IEEE.STD_LOGIC_unsigned.all;
23
24
25 entity clkdiv is
26     Port ( mclk : in  STD_LOGIC;
27           clr  : in  STD_LOGIC;
28           clk3 : out STD_LOGIC
29     );
30 end clkdiv;
31
32 architecture Behavioral of clkdiv is
33
34
35     signal q : STD_LOGIC_VECTOR(23 downto 0);
36
37
38     begin
39
40         --clock divider
41         process(mclk, clr)
42         begin
43             if clr = '1' then
44                 q <= X"000000";
45             elsif mclk'event and mclk = '1' then
46                 q <= q+1;
47             end if;
48         end process;
49
50
51         clk3 <= q(23);    --3Hz
52
53     end Behavioral;
54
55
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:      21:32:43 03/22/2011
6  -- Design Name:
7  -- Module Name:      clkdiv - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22 use IEEE.STD_LOGIC_unsigned.all;
23
24
25 entity clkdiv is
26     Port ( mclk : in  STD_LOGIC;
27           clr : in  STD_LOGIC;
28           clk190 : out STD_LOGIC
29     );
30 end clkdiv;
31
32 architecture Behavioral of clkdiv is
33
34
35     signal q : STD_LOGIC_VECTOR(23 downto 0);
36
37
38     begin
39
40         --clock divider
41         process(mclk, clr)
42         begin
43             if clr = '1' then
44                 q <= X"000000";
45             elsif mclk'event and mclk = '1' then
46                 q <= q+1;
47             end if;
48         end process;
49
50
51         clk190 <= q(18);  --190 Hz
52
53
54     end Behavioral;
55
56
```



```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    12:42:33 03/16/2011
6  -- Design Name:
7  -- Module Name:    mux21 - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
```

```
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23
24 entity mux21 is
25     Port ( a : in  STD_LOGIC;
26           b : in  STD_LOGIC;
27           s : in  STD_LOGIC;
28           y : out STD_LOGIC);
29 end mux21;
30
31 architecture Behavioral of mux21 is
32
33 begin
34     p1: process(a,b,s)
35     begin
36         if s = '0' then
37             y <= a;
38         else
39             y <= b;
40         end if;
41     end process;
42 end Behavioral;
43
44
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    16:56:40 03/23/2011
6  -- Design Name:
7  -- Module Name:    mux24 - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22 library work;
23
24 entity mux24 is
25     Port ( s : in  STD_LOGIC;
26           a : in  STD_LOGIC_VECTOR (3 downto 0);
27           b : in  STD_LOGIC_VECTOR (3 downto 0);
28           y : out STD_LOGIC_VECTOR (3 downto 0));
29 end mux24;
30
31 architecture Behavioral of mux24 is
32     component mux21
33     port(
34         a : in std_logic;
35         b : in std_logic;
36         s : in std_logic;
37         y : out std_logic
38     );
39 end component;
40
41 begin
42     U1: mux21
43     port map (a=>a(3), b=>b(3), s=>s, y=>y(3));
44     U2: mux21
45     port map (a=>a(2), b=>b(2), s=>s, y=>y(2));
46     U3: mux21
47     port map (a=>a(1), b=>b(1), s=>s, y=>y(1));
48     U4: mux21
49     port map (a=>a(0), b=>b(0), s=>s, y=>y(0));
50
51
52 end Behavioral;
53
54
```

Appendix C: Senior Defense Presentation

Senior Project VHDL Tutorials

Katherine Rodreick

Advisor: Dr. Joseph Lin

April 20, 2011



Introduction

- This project was chosen because previous projects were too time consuming for my schedule.
- Correcting examples from the lab manual and creating new tutorials to use in COIN 330.

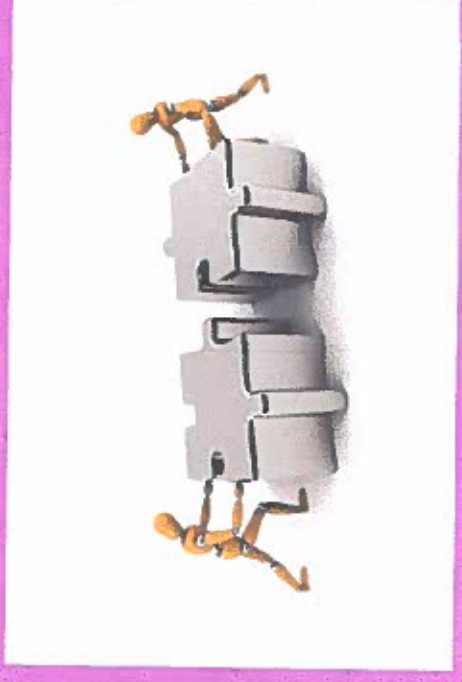
Background

- Nature of the Problem
 - The current lab manuals we use for our new FPGA board were incomplete and incorrect.
 - Most of the time confusing to read.
- Previous Solution
 - Dr. Lin began writing the tutorials by himself...
 - Problem? It was too time consuming for such a busy individual.



Background

- Proposed Solution
- Let a student make the tutorials for examples in the lab manual...
- The student ends up with a better understanding of VHDL and circuits due to the hands on and independent work they do.



Theory

- Solving the Problem...

- To solve the problem, I took up the project.

- How to Solve the Errors in the Examples.

- Trial and Error was the best theory I came up with.

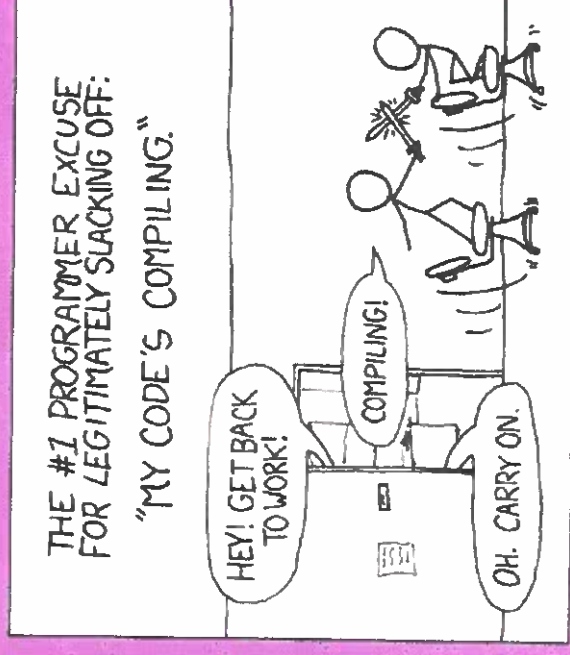
- How to Solve the Time Consumption of Writing Tutorials.

- I was not able to come up with the best solution for this besides sheer will power and the Grace of God...



Experiment

- Not much to experiment with during the process of writing the code.
- Mostly a trial and error process until figuring out what was going on during the process.
- Fun things to demonstrate.
 - Scrolling Phone Number
 - Fibonacci Example.



Conclusion

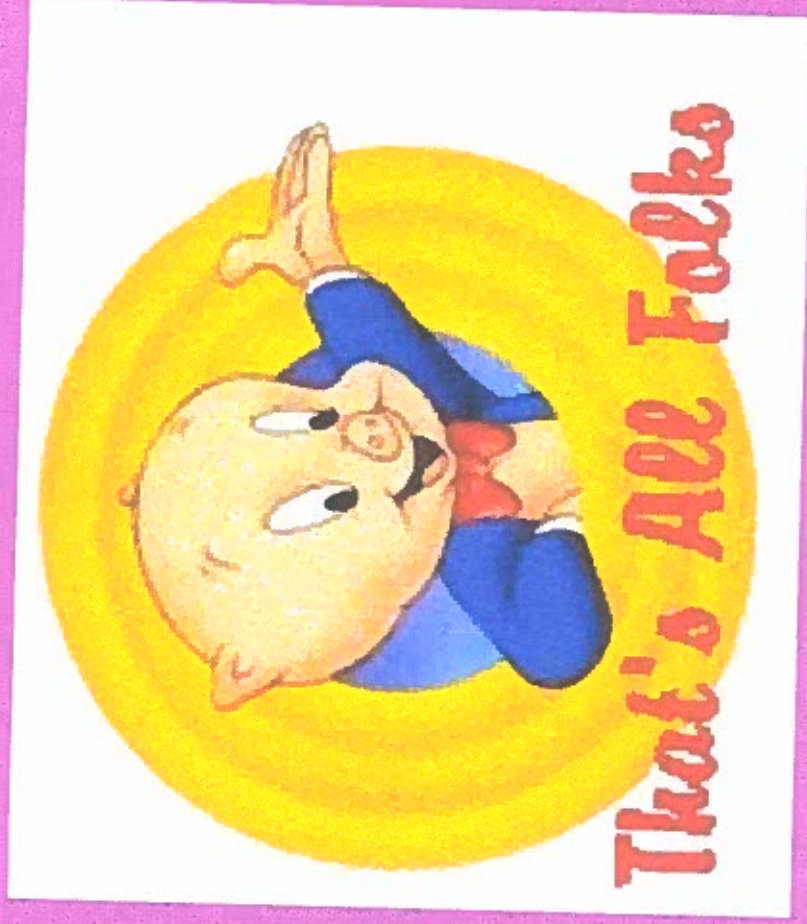
- At the end of this project I have become more efficient at building circuits from diagrams using VHDL statements.
- I use the word *let's* and *nifty* too often when trying to explain things.
- I have a better appreciation for the effort professors put into lessons.



Works Used During Project

- Haskell, Richard E. and Hanna, Darrin M.
Introduction to Digital Design Using Digilent FPGA Boards. Rochester Hills: LBE BOOKS, LLC, 2009.
Print.

The End...



hopefully...